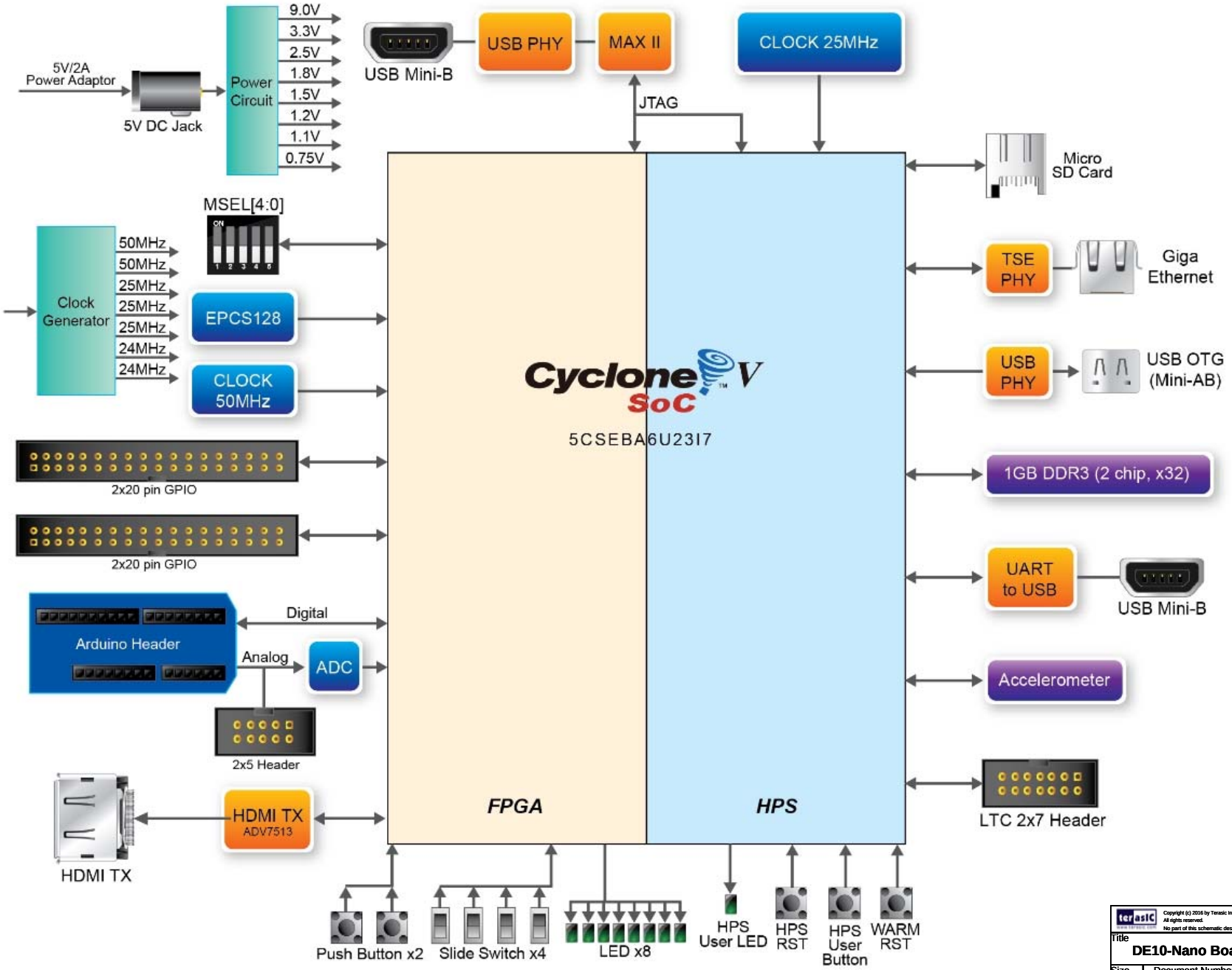

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Title
DE10-Nano Board

Size	Document Number	Rev
B	Block Diagram	A1

Date: Friday, May 05, 2017 Sheet 2 of 24

U1I

CYCLONE V SoC BANK 3

Bank 3A
VCCIO = 3.3V

GPIO_0_D32	Y11	IO_3A/PR_ERROR/DIFFIO_RX_B7P
GPIO_0_D35	AA11	IO_3A/PR_DONE/DIFFIO_RX_B7N
GPIO_0_D10	AD5	IO_3A/DIFFIO_TX_B8P/DQ1B
GPIO_0_D13	AE6	IO_3A/PR_READY/DIFFIO_TX_B8N/DQ1B

Bank 3B
VCCIO = 3.3V

HDMI_TX_D16	AE4	IO_3B/DIFFIO_TX_B9P/B_WEN/DQ2B	IO_3B/DIFFIO_TX_B17P/B_BA_0/DQ3B	AF7	GPIO_0_D6
GPIO_0_D8	AF4	IO_3B/DIFFIO_TX_B9N/GND	IO_3B/DIFFIO_TX_B17N/GND	AG6	HDMI_TX_D21
HDMI_TX_D5	AD10	IO_3B/DIFFIO_RX_B10P/B_A_14/DQ2B	IO_3B/DIFFIO_RX_B18P/B_BA_1/DQ3B	AF11	HDMI_TX_INT
HDMI_TX_D10	AE9	IO_3B/DIFFIO_RX_B10N/B_A_15/DQ2B	IO_3B/DIFFIO_RX_B18N/B_BA_2/DQ3B	AF10	HDMI_TX_D8
HDMI_LRCLK	T11	IO_3B/DIFFIO_RX_B11P/B_CSN_0/DQS2B	IO_3B/DIFFIO_RX_B19P/B_CK/DQS3B	T13	HDMI_I2S0
HDMI_MCLK	U11	IO_3B/DIFFIO_RX_B11N/B_CSN_1/DQSN2B	IO_3B/DIFFIO_RX_B19N/B_CKN/DQSN3B	T12	HDMI_SCLK
HDMI_TX_D12	AE7	IO_3B/DIFFIO_TX_B12P/B_A_12	IO_3B/DIFFIO_TX_B20P/B_A_6	AH3	GPIO_0_D9
HDMI_TX_D14	AF8	IO_3B/DIFFIO_TX_B12N/B_A_13/DQ2B	IO_3B/DIFFIO_TX_B20N/B_A_7/DQ3B	AH2	HDMI_TX_D17
HDMI_TX_D23	AE8	IO_3B/DIFFIO_TX_B13P/B_A_10/DQ2B			
HDMI_TX_D22	AF9	IO_3B/DIFFIO_TX_B13N/B_A_11/DQ2B			
HDMI_TX_D4	AD11	IO_3B/DIFFIO_RX_B14P/B_A_8/DQ2B	IO_3B/DIFFIO_RX_B22P/B_A_4/DQ3B	AE12	HDMI_TX_D1
HDMI_TX_D6	AE11	IO_3B/DIFFIO_RX_B14N/B_A_9/DQ2B	IO_3B/DIFFIO_RX_B22N/B_A_5/DQ3B	AD12	HDMI_TX_D0
HDMI_TX_D15	AF5	IO_3B/DIFFIO_TX_B16P/B_CASN/DQ2B	IO_3B/DIFFIO_TX_B24P/B_A_0/DQ3B	AH6	HDMI_TX_D20
HDMI_TX_D13	AF6	IO_3B/DIFFIO_TX_B16N/B_RASN/DQ2B	IO_3B/DIFFIO_TX_B24N/B_A_1/DQ3B	AH5	HDMI_TX_D19

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U1K

CYCLONE V SoC BANK 5

Bank 5A
VCCIO = 3.3V

LED4	AF26	IO_5A/RZQ_1/DIFFIO_TX_R1P/DQ1R			
LED5	AE26	IO_5A/PR_REQUEST/DIFFIO_TX_R1N/DQ1R			
			IO_5A/DIFFIO_RX_R6P/DQS1R	Y16	LED6
			IO_5A/DIFFIO_RX_R6N/DQSN1R	W15	LED0
			IO_5A/DIFFIO_TX_R7P/DQ1R	AA24	LED1
GPIO_1_D3	AD26	IO_5A/CVP_CONFDONE/DIFFIO_TX_R3N/DQ1R	IO_5A/DIFFIO_TX_R7N	AA23	LED7
GPIO_0_D29	Y17	IO_5A/DIFFIO_RX_R4P/DQ1R	IO_5A/DIFFIO_RX_R8P/DQ1R	V16	LED2
GPIO_0_D28	Y18	IO_5A/DIFFIO_RX_R4N/DQ1R	IO_5A/DIFFIO_RX_R8N/DQ1R	V15	LED3

Bank 5B
VCCIO = 3.3V

GPIO_0_D30	AB25	IO_5B/RZQ_2/DIFFIO_TX_R24N

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U1J

CYCLONE V SoC BANK 4

Bank 4A
VCCIO = 3.3V

Arduino_Reset_n	AH7	IO_4A/RZQ_0/DIFFIO_TX_B25N			
Arduino_IO6	AG8	IO_4A/DIFFIO_TX_B25P/B_DQ_2/DQ4B	IO_4A/DIFFIO_TX_B41P/B_DQ_18/DQ6B	AF18	GPIO_1_D30
Arduino_IO0	AG13	IO_4A/DIFFIO_RX_B26P/B_DQ_1/DQ4B	IO_4A/DIFFIO_RX_B42P/B_DQ_17/DQ6B	AE19	GPIO_1_D34
Arduino_IO1	AF13	IO_4A/DIFFIO_RX_B26N/B_DQ_0/DQ4B	IO_4A/DIFFIO_RX_B42N/B_DQ_16/DQ6B	AD19	HDMI_TX_DE
Arduino_IO4	U14	IO_4A/DIFFIO_RX_B27P/B_DQS_0/DQS4B	IO_4A/DIFFIO_RX_B43P/B_DQS_2/DQS6B	AA19	GPIO_0_D24
Arduino_IO5	U13	IO_4A/DIFFIO_RX_B27N/B_DQSN_0/DQSN4B	IO_4A/DIFFIO_RX_B43N/B_DQSN_2/DQSN6B	AA18	GPIO_0_D26
Arduino_IO3	AG9	IO_4A/DIFFIO_TX_B28P/B_ODT_0	IO_4A/DIFFIO_TX_B44P/B_RESETN	AG18	GPIO_1_D28
Arduino_IO7	AH8	IO_4A/DIFFIO_TX_B28N/B_DQ_3/DQ4B	IO_4A/DIFFIO_TX_B44N/B_DQ_19/DQ6B	AH18	GPIO_1_D29
Arduino_IO2	AG10	IO_4A/DIFFIO_TX_B29P/B_DQ_6/DQ4B	IO_4A/DIFFIO_TX_B45P/B_DQ_22/DQ6B	AG19	GPIO_1_D26
Arduino_IO14	AH9	IO_4A/DIFFIO_TX_B29N/B_ODT_1/DQ4B	IO_4A/DIFFIO_TX_B45N/GND/DQ6B	AH19	GPIO_1_D27
Arduino_IO10	AF15	IO_4A/DIFFIO_RX_B30P/B_DQ_5/DQ4B	IO_4A/DIFFIO_RX_B46P/B_DQ_21/DQ6B	AE20	GPIO_1_D33
Arduino_IO9	AE15	IO_4A/DIFFIO_RX_B30N/B_DQ_4/DQ4B	IO_4A/DIFFIO_RX_B46N/B_DQ_20/DQ6B	AD20	GPIO_0_D17
Arduino_IO15	AG11	IO_4A/DIFFIO_TX_B32P/B_DM_0/DQ4B	IO_4A/DIFFIO_TX_B48P/B_DM_2/DQ6B	AF20	GPIO_1_D31
Arduino_IO12	AH11	IO_4A/DIFFIO_TX_B32N/B_DQ_7/DQ4B	IO_4A/DIFFIO_TX_B48N/B_DQ_23/DQ6B	AG20	GPIO_1_D24
Arduino_IO13	AH12	IO_4A/DIFFIO_TX_B33P/B_DQ_10/DQ5B	IO_4A/DIFFIO_TX_B49P/B_DQ_26/DQ7B	AG21	GPIO_1_D19
Arduino_IO8	AF17	IO_4A/DIFFIO_RX_B34P/B_DQ_9/DQ5B	IO_4A/DIFFIO_RX_B50P/B_DQ_25/DQ7B	AF22	GPIO_1_D22
Arduino_IO11	AG16	IO_4A/DIFFIO_RX_B34N/B_DQ_8/DQ5B	IO_4A/DIFFIO_RX_B50N/B_DQ_24/DQ7B	AF21	GPIO_1_D25
GPIO_0_D27	W14	IO_4A/DIFFIO_RX_B35P/B_DQS_1/DQS5B	IO_4A/DIFFIO_RX_B51P/B_DQS_3/DQS7B	AD23	GPIO_0_D14
HDMI_TX_VS	V13	IO_4A/DIFFIO_RX_B35N/B_DQSN_1/DQSN5B	IO_4A/DIFFIO_RX_B51N/B_DQSN_3/DQSN7B	AE22	GPIO_1_D23
GPIO_0_D11	AG14	IO_4A/DIFFIO_TX_B36P/B_CKE_1			
GPIO_0_D5	AH13	IO_4A/DIFFIO_TX_B36N/B_DQ_11/DQ5B	IO_4A/DIFFIO_TX_B52N/B_DQ_27/DQ7B	AH21	GPIO_1_D18
GPIO_1_D32	AG15	IO_4A/DIFFIO_TX_B37P/B_DQ_14/DQ5B	IO_4A/DIFFIO_TX_B53P/B_DQ_30/DQ7B	AH23	GPIO_1_D20
GPIO_0_D7	AH14	IO_4A/DIFFIO_TX_B37N/B_CKE_0/DQ5B	IO_4A/DIFFIO_TX_B53N/GND/DQ7B	AH22	GPIO_1_D17
GPIO_0_D19	AD17	IO_4A/DIFFIO_RX_B38P/B_DQ_13/DQ5B	IO_4A/DIFFIO_RX_B54P/B_DQ_29/DQ7B	AG23	GPIO_1_D14
GPIO_1_D35	AE17	IO_4A/DIFFIO_RX_B38N/B_DQ_12/DQ5B	IO_4A/DIFFIO_RX_B54N/B_DQ_28/DQ7B	AF23	GPIO_1_D15
KEY0	AH17	IO_4A/DIFFIO_TX_B40P/B_DM_1/DQ5B	IO_4A/DIFFIO_TX_B56P/B_DM_3/DQ7B	AG24	GPIO_1_D16
KEY1	AH16	IO_4A/DIFFIO_TX_B40N/B_DQ_15/DQ5B	IO_4A/DIFFIO_TX_B56N/B_DQ_31/DQ7B	AH24	GPIO_1_D12
			IO_4A/DIFFIO_TX_B57P/B_DQ_34/DQ8B	AG26	GPIO_1_D8
			IO_4A/DIFFIO_RX_B58P/B_DQ_33/DQ8B	AE24	GPIO_0_D15
			IO_4A/DIFFIO_RX_B58N/B_DQ_32/DQ8B	AE23	GPIO_0_D12
			IO_4A/DIFFIO_RX_B59P/B_DQS_4/DQS8B	AC22	GPIO_0_D21
			IO_4A/DIFFIO_RX_B59N/B_DQSN_4/DQSN8B	AC23	GPIO_0_D20
			IO_4A/DIFFIO_TX_B60N/B_DQ_35/DQ8B	AH26	GPIO_1_D11
			IO_4A/DIFFIO_TX_B61P/B_DQ_38/DQ8B	AG28	GPIO_1_D4
			IO_4A/DIFFIO_TX_B61N/GND/DQ8B	AH27	GPIO_1_D9
			IO_4A/DIFFIO_RX_B62P/B_DQ_37/DQ8B	AF25	GPIO_1_D13
			IO_4A/DIFFIO_RX_B62N/B_DQ_36/DQ8B	AG25	GPIO_1_D10
			IO_4A/DIFFIO_TX_B64P/B_DM_4/DQ8B	AF27	GPIO_1_D7
			IO_4A/DIFFIO_TX_B64N/B_DQ_39/DQ8B	AF28	GPIO_1_D5

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User Interface (FPGA)

20	KEY[1..0]	
20	LED[7..0]	

GPIO

6,7,19	GPIO_0_D[35..0]	
6,19	GPIO_1_D[35..0]	

Arduino Digital Interface

19	Arduino_IO[15..0]	
19	Arduino_Reset_n	

HDMI TX

6,21	HDMI_TX_D[23..0]	
6,21	HDMI_TX_CLK	
6,21	HDMI_TX_HS	
6,21	HDMI_TX_VS	
6,21	HDMI_TX_DE	

HDMI Audio Interface

21	HDMI_MCLK	
21	HDMI_LRCLK	
21	HDMI_I2S0	
21	HDMI_SCLK	

HDMI Interface

21	HDMI_TX_INT	
21	HDMI_TX_DE	
21	HDMI_TX_VS	

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Title
DE10-Nano Board

Size	Document Number	Rev
B	FPGA Bank 3, 4, 5, 8	A1

Date: Friday, May 05, 2017 Sheet 3 of 24

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DDR3 Interface (HPS)

12	HPS_DDR3_ADDR[14..0]	
12	HPS_DDR3_BA[2..0]	
12	HPS_DDR3_DM[3..0]	
12	HPS_DDR3_DQ[31..0]	
12	HPS_DDR3_DQS_P[3..0]	
12	HPS_DDR3_DQS_N[3..0]	
12	HPS_DDR3_RESET_N	
12	HPS_DDR3_CK_P	
12	HPS_DDR3_CK_N	
12	HPS_DDR3_CAS_N	
12	HPS_DDR3_RAS_N	
12	HPS_DDR3_CS_N	
12	HPS_DDR3_WE_N	
12	HPS_DDR3_ODT	
12	HPS_DDR3_CKE	
12	HPS_DDR3_CS1_N	
12	HPS_DDR3_CKE1	
12	HPS_DDR3_ODT1	

U1L

CYCLONE V SoC BANK 6 (HPS)

Bank 6A
VCCIO = 1.5V

R117 100	HPS_DDR3_RZQ	D25	HPS_RZQ_0		
	HPS_DDR3_ADDR0	C28	HPS_DDR/HPS_A_0		
	HPS_DDR3_ADDR1	B28	HPS_DDR/HPS_A_1		
	HPS_DDR3_ADDR2	E26	HPS_DDR/HPS_A_2		
	HPS_DDR3_ADDR3	D26	HPS_DDR/HPS_A_3		
	HPS_DDR3_ADDR4	J21	HPS_DDR/HPS_A_4		
	HPS_DDR3_ADDR5	J20	HPS_DDR/HPS_A_5		
	HPS_DDR3_ADDR6	C26	HPS_DDR/HPS_A_6		
	HPS_DDR3_ADDR7	B26	HPS_DDR/HPS_A_7		
	HPS_DDR3_ADDR8	F26	HPS_DDR/HPS_A_8		
	HPS_DDR3_ADDR9	F25	HPS_DDR/HPS_A_9		
	HPS_DDR3_ADDR10	A24	HPS_DDR/HPS_A_10		
	HPS_DDR3_ADDR11	B24	HPS_DDR/HPS_A_11		
	HPS_DDR3_ADDR12	D24	HPS_DDR/HPS_A_12		
	HPS_DDR3_ADDR13	C24	HPS_DDR/HPS_A_13		
	HPS_DDR3_ADDR14	G23	HPS_DDR/HPS_A_14		
		F24	HPS_DDR/HPS_A_15		
	HPS_DDR3_CK_P	N21	HPS_DDR/HPS_CK		
	HPS_DDR3_CK_N	N20	HPS_DDR/HPS_CKn		
	HPS_DDR3_CKE	L28	HPS_DDR/HPS_CKE_0		
	HPS_DDR3_CKE1	K28	HPS_DDR/HPS_CKE_1		
	HPS_DDR3_BA0	A27	HPS_DDR/HPS_BA_0		
	HPS_DDR3_BA1	H25	HPS_DDR/HPS_BA_1		
	HPS_DDR3_BA2	G25	HPS_DDR/HPS_BA_2		
	HPS_DDR3_RAS_N	A25	HPS_DDR/HPS_RASn		
	HPS_DDR3_CAS_N	A26	HPS_DDR/HPS_CASn		
	HPS_DDR3_WE_N	E25	HPS_DDR/HPS_WEn		
	HPS_DDR3_CS_N	L21	HPS_DDR/HPS_CSn_0		
	HPS_DDR3_CS1_N	L20	HPS_DDR/HPS_CSn_1		
	HPS_DDR3_ODT	D28	HPS_DDR/HPS_ODT_0		
	HPS_DDR3_ODT1	G26	HPS_DDR/HPS_ODT_1		
			HPS_DDR/HPS_DQ_0	J25	HPS_DDR3_DQ0
			HPS_DDR/HPS_DQ_1	J24	HPS_DDR3_DQ1
			HPS_DDR/HPS_DQ_2	E28	HPS_DDR3_DQ2
			HPS_DDR/HPS_DQ_3	D27	HPS_DDR3_DQ3
			HPS_DDR/HPS_DQ_4	J26	HPS_DDR3_DQ4
			HPS_DDR/HPS_DQ_5	K26	HPS_DDR3_DQ5
			HPS_DDR/HPS_DQ_6	G27	HPS_DDR3_DQ6
			HPS_DDR/HPS_DQ_7	F28	HPS_DDR3_DQ7
			HPS_DDR/HPS_DQS_0	R17	HPS_DDR3_DQS_P0
			HPS_DDR/HPS_DQSn_0	R16	HPS_DDR3_DQS_N0
			HPS_DDR/HPS_DM_0	G28	HPS_DDR3_DM0
			HPS_DDR/HPS_DQ_8	K25	HPS_DDR3_DQ8
			HPS_DDR/HPS_DQ_9	L25	HPS_DDR3_DQ9
			HPS_DDR/HPS_DQ_10	J27	HPS_DDR3_DQ10
			HPS_DDR/HPS_DQ_11	J28	HPS_DDR3_DQ11
			HPS_DDR/HPS_DQ_12	M27	HPS_DDR3_DQ12
			HPS_DDR/HPS_DQ_13	M26	HPS_DDR3_DQ13
			HPS_DDR/HPS_DQ_14	M28	HPS_DDR3_DQ14
			HPS_DDR/HPS_DQ_15	N28	HPS_DDR3_DQ15
			HPS_DDR/HPS_DQS_1	R19	HPS_DDR3_DQS_P1
			HPS_DDR/HPS_DQSn_1	R18	HPS_DDR3_DQS_N1
			HPS_DDR/HPS_DM_1	P28	HPS_DDR3_DM1
			HPS_GPI10	U15	
			HPS_GPI11	U16	
			HPS_GPI12	AC27	
			HPS_GPI13	V24	

Bank 6B
VCCIO = 1.5V

	HPS_DDR3_DQ16	N24	HPS_DDR/HPS_DQ_16		
	HPS_DDR3_DQ17	N25	HPS_DDR/HPS_DQ_17		
	HPS_DDR3_DQ18	T28	HPS_DDR/HPS_DQ_18		
	HPS_DDR3_DQ19	U28	HPS_DDR/HPS_DQ_19		
	HPS_DDR3_DQ20	N26	HPS_DDR/HPS_DQ_20		
	HPS_DDR3_DQ21	N27	HPS_DDR/HPS_DQ_21		
	HPS_DDR3_DQ22	R27	HPS_DDR/HPS_DQ_22		
	HPS_DDR3_DQ23	V27	HPS_DDR/HPS_DQ_23		
	HPS_DDR3_DQS_P2	T19	HPS_DDR/HPS_DQS_2		
	HPS_DDR3_DQS_N2	T18	HPS_DDR/HPS_DQSn_2		
	HPS_DDR3_DM2	W28	HPS_DDR/HPS_DM_2		
	HPS_DDR3_DQ24	R26	HPS_DDR/HPS_DQ_24		
	HPS_DDR3_DQ25	R25	HPS_DDR/HPS_DQ_25		
	HPS_DDR3_DQ26	AA28	HPS_DDR/HPS_DQ_26		
	HPS_DDR3_DQ27	W26	HPS_DDR/HPS_DQ_27		
	HPS_DDR3_DQ28	R24	HPS_DDR/HPS_DQ_28		
	HPS_DDR3_DQ29	T24	HPS_DDR/HPS_DQ_29		
	HPS_DDR3_DQ30	Y27	HPS_DDR/HPS_DQ_30		
	HPS_DDR3_DQ31	AA27	HPS_DDR/HPS_DQ_31		
	HPS_DDR3_DQS_P3	U19	HPS_DDR/HPS_DQS_3		
	HPS_DDR3_DQS_N3	T20	HPS_DDR/HPS_DQSn_3		
	HPS_DDR3_DM3	AB28	HPS_DDR/HPS_DM_3		
			HPS_DDR/HPS_DQ_32	T26	
			HPS_DDR/HPS_DQ_33	U25	
			HPS_DDR/HPS_DQ_34	AC28	
			HPS_DDR/HPS_DQ_35	V25	
			HPS_DDR/HPS_DQ_36	V19	
			HPS_DDR/HPS_DQ_37	V20	
			HPS_DDR/HPS_DQ_38	AE27	
			HPS_DDR/HPS_DQ_39	AD28	
			HPS_DDR/HPS_DQS_4	V18	
			HPS_DDR/HPS_DQSn_4	V17	
			HPS_DDR/HPS_DM_4	AE28	
			HPS_GPI0	M25	
			HPS_GPI1	K27	
			HPS_GPI2	R20	
			HPS_GPI3	R21	
			HPS_GPI4	R28	
			HPS_GPI5	P26	
			HPS_GPI6	T17	
			HPS_GPI7	T16	
			HPS_GPI8	Y28	
			HPS_GPI9	Y26	
			HPS_DDR/HPS_RESETn	V28	HPS_DDR3_RESET_N

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Title
DE10-Nano Board

Size	Document Number	Rev
B	FPGA Bank 6	A1

Date: Friday, May 05, 2017 Sheet 4 of 24

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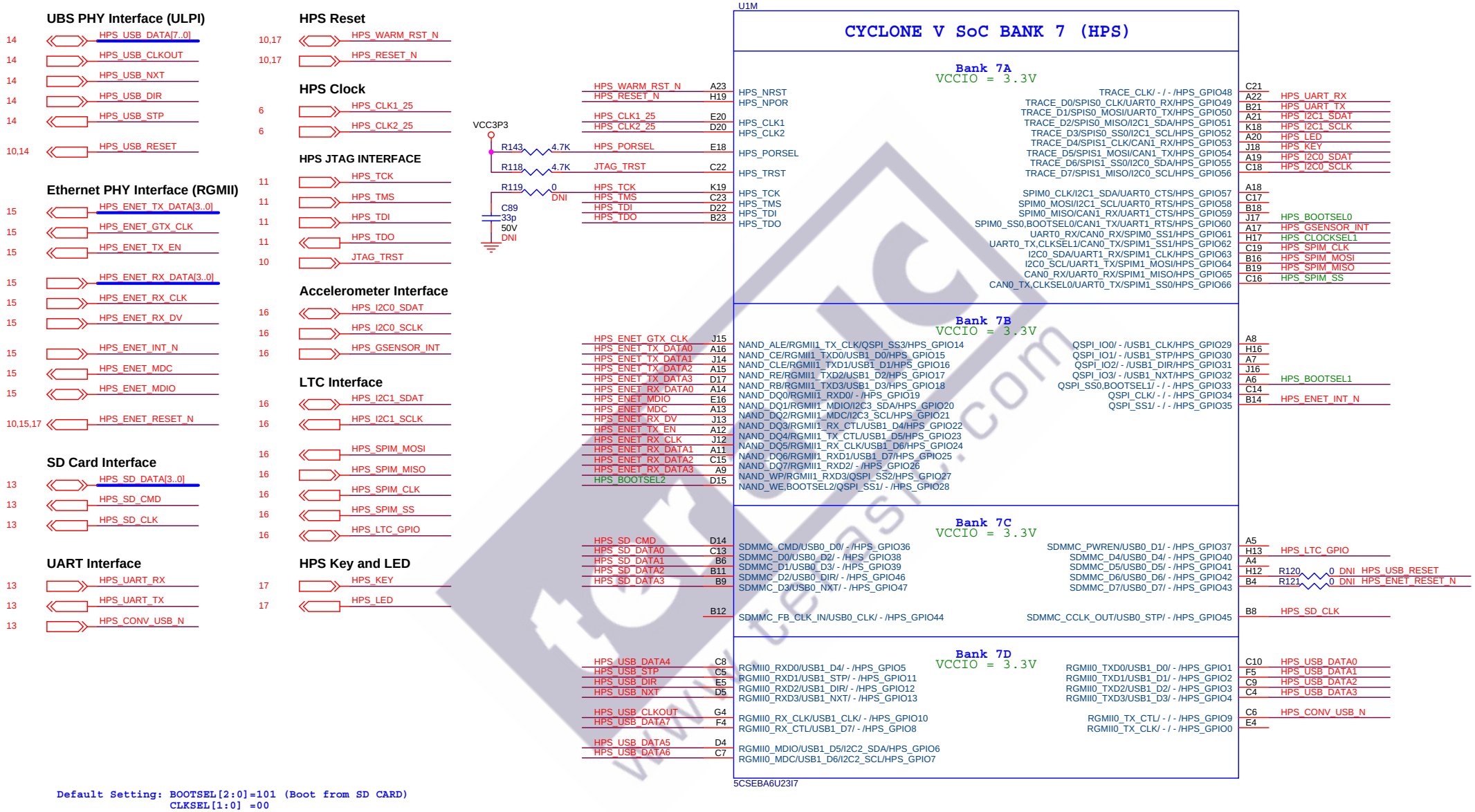

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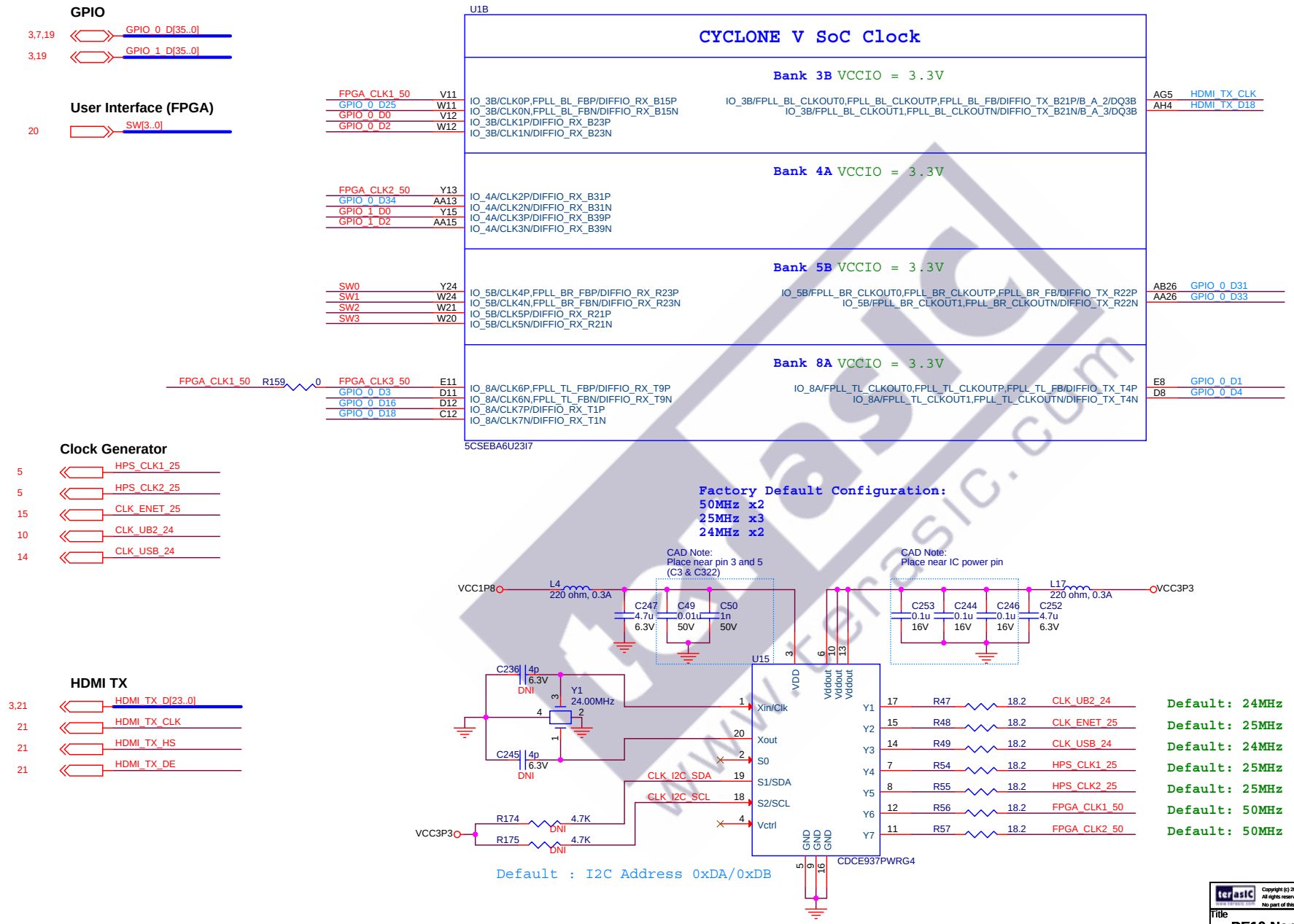
Title
DE10-Nano Board

Size	Document Number	Rev
B	FPGA Bank 7	A1

Date: Friday, May 05, 2017 Sheet 5 of 24



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Title
DE10-Nano Board

Size	Document Number	Rev
B	FPGA Clock and Clock Generator	A1

Date: Friday, May 05, 2017 Sheet 6 of 24

USB Blaster

10 JTAG_Blaster_TCK
10 JTAG_Blaster_TMS
10 JTAG_Blaster_TDO
10 JTAG_Blaster_TDI

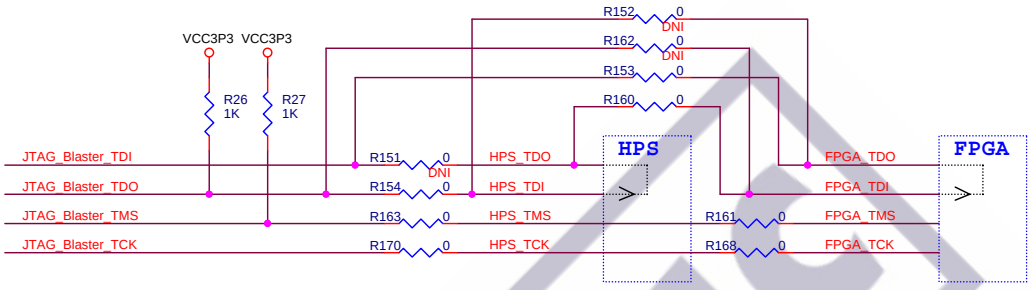
FPGA JTAG INTERFACE

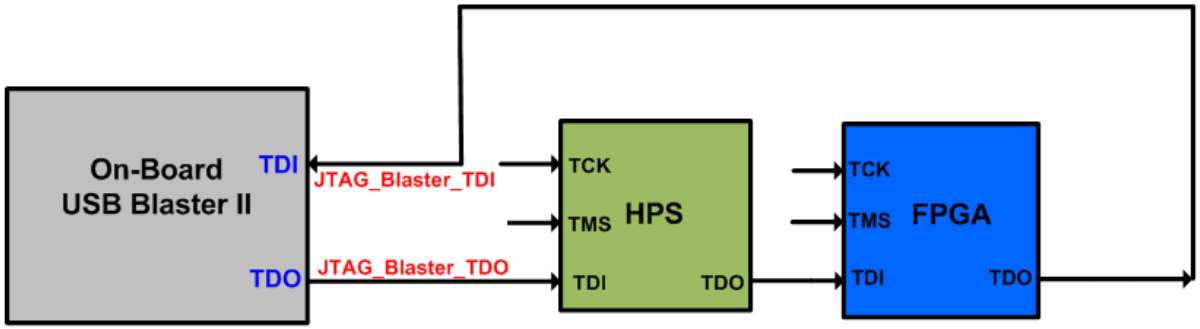
7 FPGA_TCK
7 FPGA_TMS
7 FPGA_TDI
7 FPGA_TDO

HPS JTAG INTERFACE

5 HPS_TCK
5 HPS_TMS
5 HPS_TDI
5 HPS_TDO

JTAG Chain





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Title
DE10-Nano Board

Size	Document Number	Rev
B	JTAG Chain	A1

Date: Friday, May 05, 2017 Sheet 11 of 24

GPIO

3,6,7 — GPIO_0_D[35..0]

3,6 — GPIO_1_D[35..0]

Arduino Digital Interface

3 — Arduino_IO[15..0]

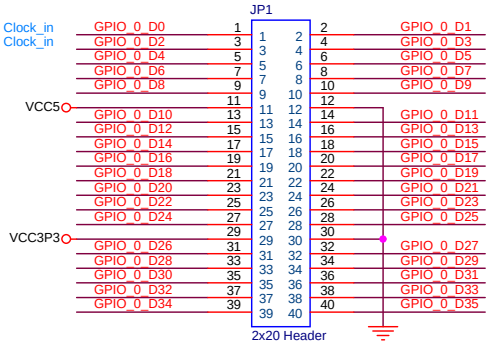
3 — Arduino_Reset_n

Analog input interface

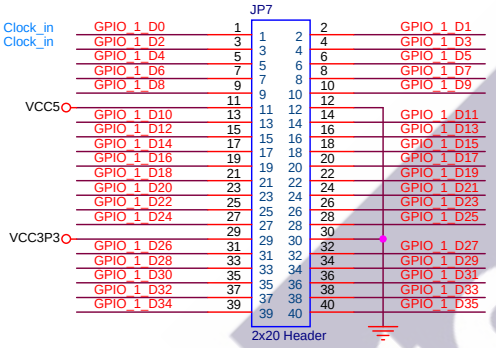
18 — ADC_IN[7..0]

18 — Arduino_VREF

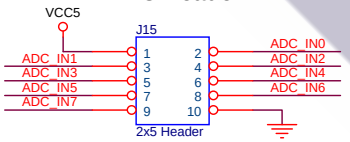
GPIO 0 Header

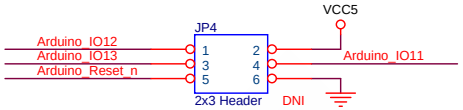


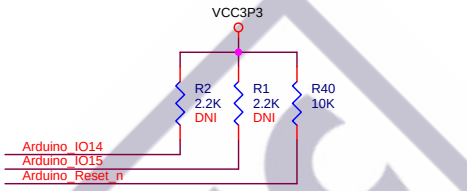
GPIO 1 Header



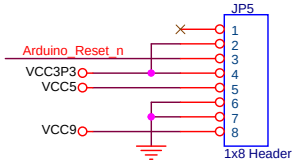
ADC Header

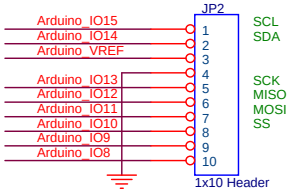


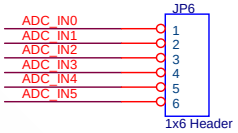


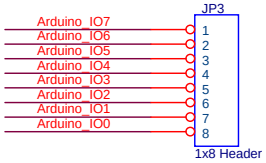


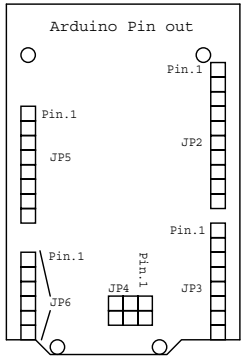
Arduino UNO Rev3











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Title
DE10-Nano Board

Size	Document Number	Rev
B	FPGA : GPIO, ADC & Arduino Expansion Header	A1

Date: Friday, May 05, 2017 Sheet 19 of 24

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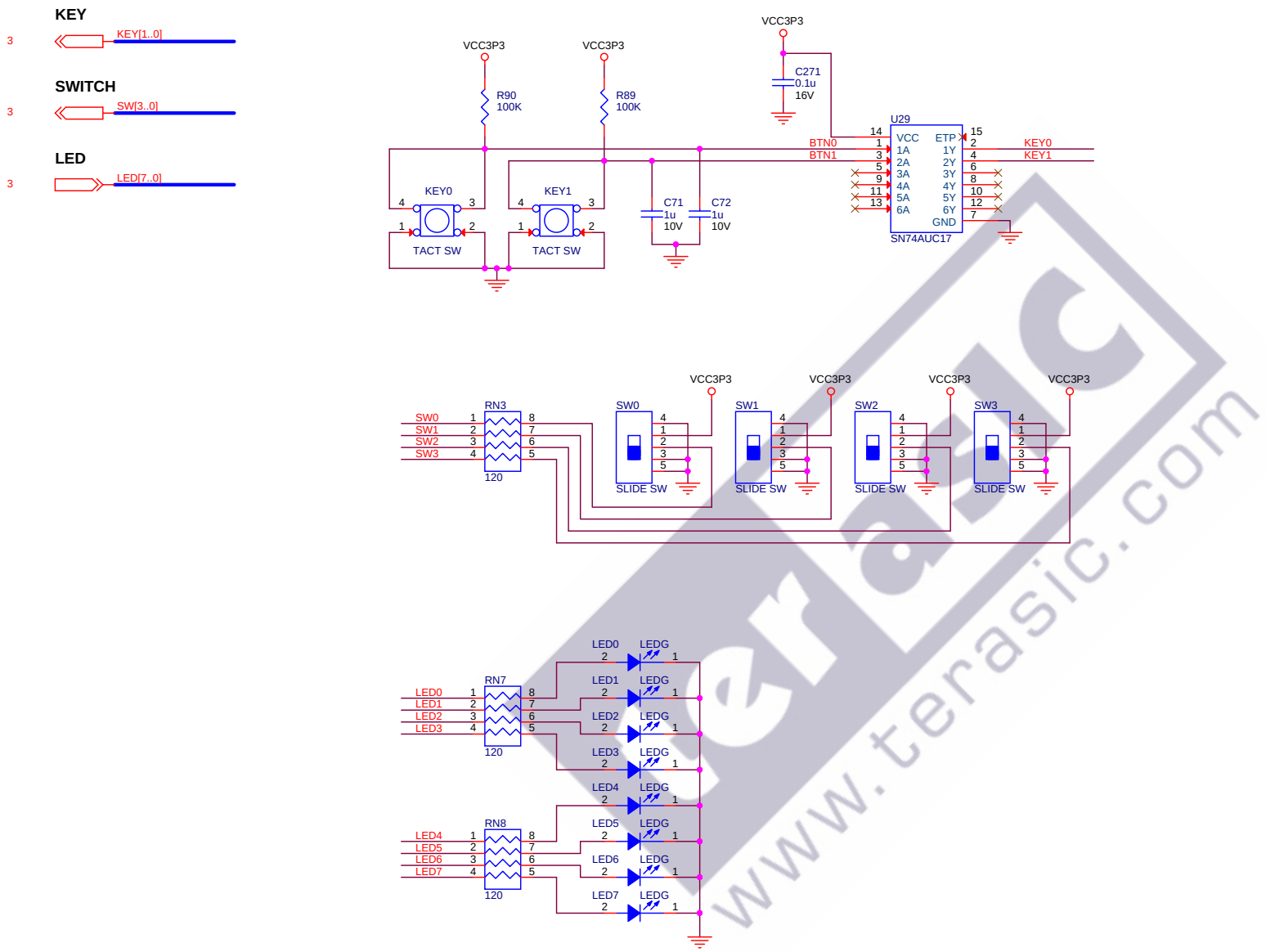

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Title
DE10-Nano Board

Size	Document Number	Rev
B	FPGA : LED, KEY, SW	A1

Date: Friday, May 05, 2017 Sheet 20 of 24